QUESTION-20

MERGE SORT

```c
#include <stdio.h>
void merge(int a[], int l, int m, int r)
{
    int i, j, k;int n1 = m - l + 1;int n2 = r - m;int L[n1], R[n2];
    for(i = 0; i < n1; i++)
        L[i] = a[l + i];
    for(j = 0; j < n2; j++)
        R[j] = a[m + 1 + j];
    i = 0;j = 0;k = l;
    while(i < n1 && j < n2)
    {
        if(L[i] <= R[j])
            a[k++] = L[i++];
        else
            a[k++] = R[j++];
    }
    while(i < n1)
        a[k++] = L[i++];

    while(j < n2)
        a[k++] = R[j++];
}
```

```
Sorted Array: 3 9 10 27 38 43 82

=== Code Execution Successful ===
```

```c
    while(j < n2)
        a[k++] = R[j++];
}
void mergeSort(int a[], int l, int r)
{
    if(l < r)
    {
        int m = (l + r) / 2;
        mergeSort(a, l, m);
        mergeSort(a, m + 1, r);
        merge(a, l, m, r);
    }
}
int main()
{
    int a[] = {38, 27, 43, 3, 9, 82, 10};
    int n = 7, i;
    mergeSort(a, 0, n - 1);
    printf("Sorted Array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}
```


Sorted Array: 3 9 10 27 38 43 82

=== Code Execution Successful ===